Closed loop control of three-phase AC-DC converter-III

Introduction

In the rapidly evolving landscape of charging infrastructure, efficient and reliable power conversion is paramount. Three-phase AC-DC converters are foundational components in these systems, transforming alternating current from the grid into direct current suitable for charging electric vehicles or powering various DC loads. However, simply converting AC to DC is not enough; modern applications demand precise control over the output voltage and a harmonious interaction with the grid, specifically by drawing current with a unity power factor. This ensures minimal reactive power consumption and optimal grid efficiency.

This lecture delves into the sophisticated closed-loop control strategies essential for achieving these critical objectives in three-phase AC-DC converters. We explore how advanced control techniques, such as reference frame transformations and feedforward compensation, enable independent regulation of output voltage and power factor. Understanding these mechanisms is crucial for designing robust and high-performance charging systems that meet the stringent demands of today's electrical grids.

Core Concepts

Term	Definition	Significance
`Unity Power Factor (UPF)`	A condition where the phase angle between the AC source voltage and current is zero, meaning current and voltage are in phase.	Minimizes reactive power drawn from the grid, improving efficiency and reducing grid stress.
`DQ Frame`	A synchronous rotating reference frame (direct and quadrature axes) used to transform three-phase AC quantities into two mutually independent DC quantities.	Simplifies control by converting time-varying AC signals into constant DC values, allowing the use of conventional PI controllers.

Term	Definition	Significance
`Cross-Coupling Terms`	Interdependencies between the D-axis and Q-axis components in the DQ frame model of the converter.	If not compensated, these terms make independent control of D and Q components difficult, hindering precise regulation.
`PI Controller`	A Proportional-Integral controller, a common feedback control loop mechanism that calculates an error value as the difference between a measured process variable and a desired setpoint.	Effective for controlling DC quantities in the DQ frame due to its ability to provide infinite gain at DC (s=0), ensuring zero steady-state error.
`Feedforward Control`	A control strategy that anticipates and compensates for disturbances or known system dynamics before they affect the output.	Used to cancel out cross-coupling terms and known voltage components in the DQ model, decoupling the D and Q axes for independent control.
`Average Time Delay (TD)`	The inherent delay in the response of a pulse-width modulated (PWM) converter, typically approximated as half of the switching period (TSW/2).	Must be accounted for in controller design to maintain stability and performance, as it introduces phase lag into the control loop.
`Space Vector`	A complex number representation that combines the instantaneous values of three-phase quantities (e.g., voltages or currents) into a single rotating vector.	Provides a unified representation for analyzing and controlling three-phase systems, simplifying transformations between reference frames.

Detailed Analysis

1. Control Objectives and Reference Frame Transformation

The primary objectives for controlling a three-phase AC-DC converter are twofold:
* **Regulating Output Voltage**: Maintaining the DC output voltage (V0) at a desired setpoint, regardless of load variations or input disturbances.
* **Achieving Unity Power Factor (UPF)**: Ensuring that the current drawn from the AC source is in phase with the supply voltage, minimizing reactive power exchange with the grid. This implies that the phase angle between the phase current and phase voltage is zero.

To achieve these objectives, especially independent control, a series of transformations are applied to the three-phase quantities:
* **Three-Phase to Two-Phase (Alpha-Beta) Transformation**: The initial step converts the three interdependent phase quantities (ISA, ISB, ISC) into two mutually independent stationary quantities (IS alpha, IS beta). This is crucial because in a balanced three-phase system, the sum of currents is zero (ISA + ISB + ISC = 0), meaning they are not independently controllable. The alpha and beta axes are 90 degrees apart, ensuring mutual independence.
* **Two-Phase (Alpha-Beta) to DQ (Synchronous Rotating) Transformation**: The alpha-beta quantities, while independent, still vary sinusoidally with time. To simplify control, they are further transformed into the synchronous rotating DQ frame. The key here is to align the direct (D) axis with the supply voltage space vector (Vs). This alignment is achieved by ensuring the quadrature (Q) component of the supply voltage (VSQ) is zero.
* When the D-axis is aligned with Vs, the Vs space vector only has a VSD component, and VSQ is zero.
* In the DQ frame, all three space vectors (Vs, IS, Vconverter) rotate at the same angular speed (omega) as the DQ frame itself. Consequently, when viewed from the DQ frame, these quantities appear as stationary, or DC, values. This is a critical advantage, as DC quantities are much easier to control using conventional PI controllers.
* **Unity Power Factor Realization**: To achieve UPF, the current space vector (IS) must be aligned with the voltage space vector (Vs). Since the D-axis is already aligned with Vs, this means the IS space vector must also be aligned with the D-axis. This condition is met by setting the quadrature component of the current (ISQ) to zero.
* If ISQ = 0, then the IS space vector consists solely of the ISD component (IS = ISD + j0).
* The magnitude of the current drawn from the grid can then be controlled directly by adjusting the ISD component. This provides independent control over active

power (related to ISD) and reactive power (related to ISQ, which is set to zero for UPF).

2. DQ Axis Model and Cross-Coupling

The dynamics of the converter in the DQ frame can be modeled by two coupled differential equations for the D and Q axis currents. These equations include terms representing the inductor (L), series resistance (RS), converter voltages (Vconverter D, Vconverter Q), and importantly, cross-coupling terms:
* **D-axis model**: L (dISD/dt) + RS ISD = -Vconverter D + VSD - omega L ISQ
* **Q-axis model**: L (dISQ/dt) + RS ISQ = -Vconverter Q + VSQ + omega L ISD

The terms `omega L ISQ` in the D-axis model and `omega L ISD` in the Q-axis model are the **cross-coupling terms**. They indicate that a change in the Q-axis current affects the D-axis dynamics, and vice-versa. This coupling makes independent control of ISD and ISQ challenging. Additionally, the `VSD` and `VSQ` terms represent the supply voltage components, which are external disturbances to the current control loop.

3. Converter Dynamics and Delay

The three-phase half-bridge converter, typically modulated using Sinusoidal Pulse Width Modulation (SPWM), introduces its own dynamics and delays:
* **Modulation**: SPWM ensures that the average pole voltages at the converter output vary sinusoidally, allowing for smooth AC current control. The modulating wave (Vm) is compared against a high-frequency carrier wave (Fc or FSW).
* **Average Time Delay (TD)**: Due to the discrete nature of PWM switching, there's an inherent delay between a change in the modulating wave and its reflection in the average output voltage of the converter.
* The best-case scenario for this delay is 0 (change occurs at the start of a switching cycle).
* The worst-case scenario is one full switching cycle (TSW) (change occurs just after a comparison, requiring waiting for the next cycle).
* The **average delay** is approximated as TSW/2, where TSW = 1/FSW (switching period).
* **Converter Transfer Function**: For control purposes, the three-phase half-bridge converter can be modeled as a simple gain (G) with an average time delay. Its transfer function is given by:
G / (1 + sTD)
Where G is the gain (e.g., V0/2Vc, relating output voltage to modulating signal) and TD = TSW/2.

4. Feedforward Control for Decoupling

To overcome the challenges posed by cross-coupling terms and supply voltage components, feedforward control is employed:
* **Purpose**: Feedforward terms are added to the control signals (Vconverter D, Vconverter Q) to actively cancel out the undesirable terms in the DQ axis models.
* **D-axis Feedforward**: To cancel `VSD - omega L ISQ`, the feedforward term `-(VSD - omega L ISQ)` is added to the Vconverter D command. When divided by the converter gain G, this becomes `(omega L ISQ - VSD) / G`.
* **Q-axis Feedforward**: To cancel `VSQ + omega L ISD`, the feedforward term `-(VSQ + omega L ISD)` is added to the Vconverter Q command. When divided by G, this becomes `(-omega L ISD - VSQ) / G`.
* **Result**: With effective feedforward compensation, the D-axis and Q-axis current dynamics become completely decoupled, allowing independent control of ISD and ISQ. This simplifies the control problem significantly, as each axis can be treated as a separate, single-input, single-output system.

5. Closed-Loop Control Block Diagram

The overall closed-loop control system for the three-phase AC-DC converter typically consists of an outer voltage control loop and inner current control loops:

- **Outer Voltage Control Loop**:
- The desired output DC voltage (V0_reference) is compared with the actual measured output voltage (V0_feedback).
- The error signal is fed into a **Voltage Controller** (typically a PI controller).
- The output of the voltage controller, after passing through a limiter, generates the **ISD_reference**. This is because the active power drawn from the source (which determines the output voltage) is primarily controlled by the D-axis current component (ISD) when operating at UPF.
- The voltage feedback is obtained via a voltage sensor with a transfer function K2 / (1 + sT2).
- **Inner Current Control Loops (D-axis and Q-axis)**:
- **D-axis Current Loop**:
- The ISD_reference (from the voltage controller) is compared with the actual measured D-axis current (ISD_feedback).
- The error is fed into an **ISD Current Controller** (a PI controller).
- The output of this controller is a component of the Vconverter D command.

- This is then combined with the D-axis feedforward term `(omega L ISQ - VSD) / G`.
- The combined signal drives the converter's D-axis input, which then passes through the converter transfer function (G / (1 + sTD)) and the plant dynamics (1 / (sL + RS)) to produce the ISD current.
- **Q-axis Current Loop**:
- For unity power factor operation, the **ISQ_reference is set to zero**.
- This zero reference is compared with the actual measured Q-axis current (ISQ_feedback).
- The error is fed into an **ISQ Current Controller** (a PI controller).
- The output of this controller is a component of the Vconverter Q command.
- This is then combined with the Q-axis feedforward term `(-omega L ISD - VSQ) / G`.
- The combined signal drives the converter's Q-axis input, which then passes through the converter transfer function (G / (1 + sTD)) and the plant dynamics (1 / (sL + RS)) to produce the ISQ current.
- Current feedback for both loops is obtained via current sensors with a transfer function K1 / (1 + sT1).
- **PI Controller Application**: Since the DQ quantities are DC, conventional PI controllers are highly effective. They introduce a pole at s=0, providing infinite gain at DC, which ensures that the feedback accurately tracks the reference with zero steady-state error.

This hierarchical control structure, combining outer voltage regulation with inner decoupled current control, allows for precise and independent management of both output voltage and grid power factor.

Key Takeaways

- [ ] The primary control objectives for three-phase AC-DC converters are output voltage regulation and unity power factor operation.
- [ ] Transformation from three-phase to the synchronous rotating DQ frame converts time-varying AC quantities into constant DC quantities, simplifying control.
- [ ] Aligning the D-axis with the supply voltage space vector (Vs) and setting the Q-axis current (ISQ) to zero ensures unity power factor.
- [ ] The DQ axis models of the converter contain cross-coupling terms (e.g., omega L ISQ, omega L ISD) that must be compensated for independent control.

- [ ] Feedforward control is used to cancel out cross-coupling terms and supply voltage components, effectively decoupling the D and Q axes.
- [ ] The converter introduces an average time delay (TSW/2) due to PWM switching, which needs to be considered in controller design.
- [ ] A hierarchical control structure with an outer voltage loop and inner decoupled current loops (using PI controllers) is employed for precise control.
- [ ] PI controllers are suitable for controlling the DC quantities in the DQ frame, providing high gain at DC for accurate tracking.

Conclusion

The detailed exploration of closed-loop control for three-phase AC-DC converters reveals a sophisticated interplay of reference frame transformations, dynamic modeling, and advanced control techniques. By converting complex, interdependent AC quantities into manageable DC components within the synchronous rotating DQ frame, and by meticulously compensating for cross-coupling effects through feedforward strategies, engineers can achieve independent and precise control over both the converter's output voltage and its interaction with the grid.

This capability is not merely an academic exercise; it is fundamental to the development of efficient and reliable charging infrastructure, renewable energy integration, and other power electronics applications. The ability to maintain a stable DC output while drawing current at unity power factor ensures optimal energy transfer, minimizes grid disturbances, and contributes to the overall stability and efficiency of the electrical network. As power demands continue to grow and grid requirements become more stringent, these advanced control methodologies will remain at the forefront of power electronics design, paving the way for smarter and more sustainable energy systems.